NEUROMORPHIC ECG ABNORMAL DETECTION USING FPGA

A Mini Project work submitted in partial fulfillment of the requirement for the award of the degree of

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in
ELECTRONICS & COMMUNICATION ENGINEERING

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We hereby declare that the project entitled " Neuromorphic ECG Abnormal Detection Using FPGA" submitted to B V Raju Institute of Technology , affiliated to Jawaharlal Nehru Technological University (JNTU), Hyderabad, for the degree of Bachelor of Technology (B.Tech) in Electronics and Communication Engineering (ECE) is a result of original project work done by us.

It is further declared that the project report on any part, therefore, has not been previously submitted to any University or institute for the award of a degree or diploma.

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ABSTRACT

Detecting cardiac abnormalities from electrocardiogram (ECG) signals is crucial in domains such as healthcare monitoring, clinical diagnosis, and emergency response, where precise and timely analysis is essential for informed medical decision-making. High accuracy and real-time processing of ECG signals enable physicians to detect arrhythmias accurately, monitor cardiac conditions over time, and assess critical situations effectively. Traditional ECG classification techniques based on software implementations and conventional neural networks have proven effective in improving detection accuracy. However, these methods often involve complex computations, making them less practical for real-time processing, especially in embedded systems or edge computing environments where resource constraints are a concern.

To address these limitations, this project introduces a hardware-based neuromorphic solution for ECG abnormality detection utilizing Field-Programmable Gate Arrays (FPGAs). By employing the Leaky Integrate-and-Fire (LIF) neuron model implemented in Verilog RTL on a Zynq-7020 FPGA platform along with AXI Direct Memory Access (DMA) protocol for high-speed PS-PL data transfer, a 10-neuron Spiking Neural Network (SNN) is realized in hardware for real-time ECG classification. ECG samples from the MIT-BIH Arrhythmia Database are processed through the SNN layer, where each neuron integrates the input signal and generates spike events based on membrane potential dynamics. The experimental results demonstrate that the proposed FPGA-based neuromorphic system successfully detects ECG abnormalities using spike-based computation on reconfigurable hardware, validating the feasibility of neuromorphic computing as a practical and efficient alternative for real-time biomedical signal processing applications.

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CHAPTER 1

INTRODUCTION

1.1 PROJECT OVERVIEW

The human heart generates electrical signals that regulate its rhythmic contractions, and these signals can be recorded and analyzed through electrocardiography (ECG). ECG signals provide valuable information about the cardiac health of an individual, enabling physicians to diagnose a wide range of heart conditions including arrhythmias, myocardial infarction, and ventricular fibrillation. With cardiovascular diseases remaining the leading cause of death worldwide, accurate and timely detection of Cardiac abnormalities have become a critical requirement in modern healthcare systems.

Traditional ECG analysis relies heavily on manual interpretation by trained cardiologists, which is time-consuming and prone to human error, particularly in highvolume clinical settings. Automated ECG classification systems have therefore gained significant attention as a means of improving diagnostic accuracy and reducing the burden on healthcare professionals. These systems employ various signal processing and machine learning techniques to analyze ECG waveforms and identify abnormal patterns with high precision.

Conventional machine learning approaches such as Support Vector Machines (SVM), k-Nearest Neighbors (kNN), and Decision Trees have been widely applied for ECG classification. While these methods have demonstrated reasonable accuracy, they require extensive feature engineering and are often sensitive to noise and signal variability. More recently, deep learning techniques such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks have shown superior performance in ECG classification tasks due to their ability to automatically extract relevant features from raw signals.

However, deep learning models are computationally intensive and require significant hardware resources, making them challenging to deploy in real-time embedded systems. Wearable cardiac monitoring devices and implantable sensors operate under strict power and resource constraints, necessitating the development of lightweight and energy-efficient classification algorithms. This has motivated researchers to explore alternative computing paradigms that can deliver high performance while consuming minimal power.

Neuromorphic computing has emerged as a promising solution to this challenge, drawing inspiration from the biological neural networks of the human brain. Unlike conventional artificial neural networks that use continuous-valued activations, neuromorphic systems employ Spiking Neural Networks (SNNs) that communicate through discrete spike events, like biological neurons. This event-driven nature of SNNs results in significantly lower power consumption compared to traditional neural networks, making them highly suitable for edge computing and embedded applications.

The Leaky Integrate-and-Fire (LIF) neuron model is one of the most widely used models in neuromorphic computing due to its simplicity and biological plausibility. In the LIF model, a neuron accumulates input charge in its membrane potential and generates a spike when the potential exceeds a predefined threshold. The membrane potential then resets and gradually leaks back to its resting state, mimicking the behavior of biological neurons. This simple yet effective model can be efficiently implemented in hardware, making it ideal for FPGA-based neuromorphic systems.

Field-Programmable Gate Arrays (FPGAs) have gained considerable attention as a hardware platform for implementing neuromorphic computing systems. FPGAs offer reconfigurability, parallel processing capabilities, and low power consumption, making them well-suited for real-time signal processing applications. Unlike Application-Specific Integrated Circuits (ASICs), FPGAs can be reprogrammed to accommodate design changes, providing greater flexibility during the development and testing phases.

The Xilinx Zynq-7020 System-on-Chip (SoC) combines a dual-core ARM Cortex-A9 Processing System (PS) with a Xilinx 7-series FPGA Programmable Logic (PL) fabric on a single chip. This PS-PL architecture enables efficient co-design of hardware and software components, allowing complex signal processing tasks to be offloaded from the processor to the reconfigurable logic. The AXI Direct Memory Access (DMA) protocol facilitates high-speed data transfer between the PS and PL, enabling real-time processing of ECG samples through the neuromorphic hardware accelerator.

The MIT-BIH Arrhythmia Database is a widely used benchmark dataset for ECG classification research, containing 48 half-hour recordings of ambulatory ECG signals from 47 patients. Each recording is annotated by cardiologists, providing ground truth labels for normal and abnormal beats. This database has been extensively used to evaluate the performance of automated ECG classification systems, making it an ideal choice for validating the proposed neuromorphic approach.

In this project, a hardware-based neuromorphic ECG abnormality detection system is designed and implemented on the Zynq-7020 FPGA platform. A 10-neuron SNN using the LIF neuron model is realized in Verilog RTL and integrated into the Zynq PS-PL co-design architecture. ECG samples from the MIT-BIH Arrhythmia Database are processed through the SNN layer in real time, and the aggregate spike count is used to classify each sample as normal or abnormal. The proposed system demonstrates the feasibility of neuromorphic computing for biomedical signal processing on reconfigurable hardware,

offering a practical and energy-efficient alternative to conventional deep learning approaches.

1.2 MOTIVATION

Cardiovascular diseases remain the leading cause of death worldwide, with millions of lives lost annually due to undetected cardiac abnormalities. Early and accurate detection of arrhythmia through continuous ECG monitoring can significantly reduce mortality rates. However, existing cardiac monitoring systems are largely confined to clinical environments, failing to capture transient cardiac events that occur unpredictably in everyday life. This has driven the demand for wearable and real-time ECG classification systems that can operate efficiently outside hospital settings.

Deep learning models such as CNN and LSTM have shown remarkable accuracy in ECG classification but are computationally expensive and power hungry, making them impractical for battery-powered wearable devices. There is therefore a growing need for lightweight and energy-efficient classification algorithms that can match the accuracy of deep learning approaches without the associated computational burden. Neuromorphic computing, inspired by the brain energy-efficient spike-based communication, presents a compelling alternative that consumes significantly lower power while maintaining competitive classification performance.

The Leaky Integrate-and-Fire neuron model offers a simple yet biologically plausible mechanism for processing temporal signals like ECG waveforms, where the timing and frequency of spikes naturally encode the signal dynamics. Implementing such neuromorphic systems on FPGAs enables massively

parallel hardware execution that closely mirrors the parallel nature of biological neural networks, resulting

in improved processing speed and energy efficiency compared to software implementations.

The Xilinx Zynq-7020 SoC provides an accessible and flexible platform for realizing a complete neuromorphic ECG classification system, combining the programmability of an ARM processor with the parallelism of reconfigurable FPGA logic. This project is motivated by the opportunity to bridge the gap between neuromorphic computing research and practical biomedical applications by demonstrating a fully

functional SNN-based ECG abnormality detection system on commercially available FPGA hardware.

Furthermore, the increasing prevalence of Internet of Things (IoT) devices in healthcare has created a pressing need for intelligent edge computing solutions that can perform complex signal analysis locally without relying on cloud connectivity. The proposed neuromorphic approach aligns perfectly with this requirement, as it enables ondevice processing with minimal latency and power consumption. By implementing SNN directly on FPGA hardware, the system eliminates the need for data transmission to remote servers, thereby addressing privacy concerns and reducing communication overhead in healthcare monitoring applications.

The convergence of neuromorphic computing, FPGA technology, and biomedical signal processing represents a significant research frontier with immense potential for improving patient outcomes. This project aims to contribute to this emerging field by providing a practical demonstration of neuromorphic ECG analysis on reconfigurable hardware, establishing a foundation for future advancements in wearable and implantable cardiac monitoring systems.

1.3 PROBLEM STATEMENT

Electrocardiogram (ECG) signals are widely used for detecting heart abnormalities such as arrhythmias, but accurate and real-time analysis of these signals remains a challenging task. ECG signals are often affected by noise, signal variability, and large volumes of continuous data, which makes processing and interpretation difficult. Traditional methods for ECG abnormality detection are mostly software-based and rely on complex algorithms executed on general-purpose processors. Although these methods provide reliable results, they often suffer from higher latency and are not efficient for real time applications.

In addition, conventional systems are not optimized for hardware implementation and lack the ability to process data in parallel, which is essential for continuous monitoring systems. With the increasing demand for portable and real-time healthcare devices, there is a need for a system that can perform fast and efficient ECG analysis with low power consumption. Therefore, this project focuses on developing a neuromorphic ECG abnormality detection system using FPGA, where a Spiking Neural Network (SNN) is implemented to process ECG signals efficiently. The proposed system aims to achieve Realtime performance, reduced latency, and improved efficiency through hardware-based parallel processing.

The specific challenges addressed by this project include: (1) the high computational complexity of existing deep learning-based ECG classification methods that prevents their deployment on resource-constrained edge devices; (2) the lack of hardware-optimized neuromorphic architectures for biomedical signal processing that can leverage the inherent parallelism of FPGA platforms; (3) the need for a complete end-to-end system that integrates data acquisition, neuromorphic processing, and result visualization on a single hardware platform; and (4) the requirement for a scalable architecture that can be extended to support multi-lead ECG processing and multi-class arrhythmia detection in future iterations.

1.4 OBJECTIVES

1. To design and implement a Spiking Neural Network (SNN) using the Leaky Integrate-and-Fire (LIF) neuron model in Verilog RTL on the Xilinx Zynq-7020 FPGA platform for ECG signal classification.
2. To develop a complete PS-PL co-design architecture integrating the ARM CortexA9 Processing System with the neuromorphic hardware accelerator using AXI Direct Memory Access (DMA) protocol for high-speed data transfer between the processor and the FPGA programmable logic.

3. To validate the proposed neuromorphic ECG abnormality detection system using real patient ECG signals from the MIT-BIH Arrhythmia Database and demonstrate its feasibility as an energy-efficient alternative to conventional deep learning approaches for biomedical signal processing applications.
4. To analyze the performance of the implemented system in terms of spike-based detection accuracy, processing latency, hardware resource utilization, and power efficiency.
5. To establish a scalable neuromorphic architecture that can serve as a foundation for future extensions including multi-class arrhythmia detection, adaptive on-chip learning, and integration with wearable cardiac monitoring devices.

1.5 SCOPE OF THE PROJECT

This project focuses on the design and implementation of a neuromorphic Spiking Neural Network using the Leaky Integrate-and-Fire neuron model on the PYNQ-Z2 Zynq7020 FPGA platform for ECG abnormality detection. The hardware implementation includes the development of LIF neuron RTL design in Verilog, a 10-neuron SNN architecture, and a complete PS-PL co-design system using AXI DMA protocol for data transfer between the ARM Cortex-A9 processor and the FPGA programmable logic. The MIT-BIH Arrhythmia Database is used as the primary data source for validation, where ECG samples are normalized and processed through the neuromorphic hardware to classify each sample as normal or abnormal based on a predefined spike count threshold.

The scope of this project is limited to binary classification of ECG signals and does not extend to multi-class arrhythmia detection or clinical diagnosis. The synaptic weights of the SNN are initialized with predefined values for hardware verification purposes, and the optimization of these weights through training on the MIT-BIH dataset is identified as future work beyond the current scope. Similarly, the abnormality threshold used for classification is set as a baseline value and will require further optimization following weight training to achieve improved classification accuracy.

The project does not include the design of a custom ASIC or the integration of a physical ECG sensor for live signal acquisition. The focus remains on demonstrating the feasibility of neuromorphic computing for ECG classification on reconfigurable FPGA hardware, providing a strong foundation for future extensions such as multi-lead ECG processing, on-chip learning, weight training using neuromorphic frameworks, and eventual deployment on wearable cardiac monitoring devices.

1.6 ORGANISATION OF THE REPORT

This report is systematically organized into several chapters to provide a clear understanding of the development, implementation, and evaluation of the proposed neuromorphic ECG abnormality detection system.

Chapter 1 introduces the project, discussing the motivation behind the work, the problem statement, the main objectives, the scope of the project, and a brief overview of the report structure.

Chapter 2 presents a comprehensive literature survey that covers the existing methods related to ECG signal processing, neuromorphic computing, and FPGA-based implementations, highlighting them limitations and explaining the contributions of the proposed work.

Chapter 3 deals with the analysis and design phase, which includes system architecture, design of the Spiking NeuralNetwork (SNN), modeling of the Leaky Integrate-and-Fire (LIF) neuron, and the overallhardware-software co-design approach. It also explains the FPGA architecture, AXI DMA communication, and integration of the processing system (PS) with programmable logic (PL) on the PYNQ-Z2 platform.

Chapter 4 explains the implementation of the proposed system, including RTL design of LIF neurons, weight ROM, and top module, along with the Vivado block design consisting of Zynq Processing System, AXI DMA, and interconnects. It also describes the PYNQ-based execution using Jupyter Notebook, data transfer process, and ECG signal handling from the MIT-BIH dataset.

Chapter 5 presents the experimental results obtained from processing ECG signals, including waveform visualization, spike count analysis, and abnormality detection performance. It also discusses system efficiency, real-time processing capability, and hardware performance.

Chapter 6 concludes the report by summarizing the overall outcomes and discussing the future scope for improving the system, such as weight optimization, adaptive learning, and integration with real-time healthcare monitoring systems.

CHAPTER 2

LITERATURE SURVEY

2.1 EXISTING SYSTEMS

The advancement of ECG abnormality detection has led to the development of various techniques aimed at improving accuracy, processing speed, and real-time performance. With the increasing demand for efficient healthcare systems, researchers have explored FPGA-based implementations and neuromorphic computing approaches to overcome the limitations of traditional software-based methods. Several notable research efforts have contributed to this field by introducing hardware-accelerated and biologically inspired models for biomedical signal processing. Below is an overview of existing systems and their contributions.

Li et al. developed an ECG classification system using deep learning techniques, where convolutional neural networks were used to extract features and classify cardiac signals. Their method achieved high classification accuracy; however, the computational complexity and memory requirements were significantly high, making it less suitable for real-time embedded applications. [1]

Mohammed Alareqi et al. proposed a biomedical signal processing framework using FPGA technology. Their work utilized MATLAB Simulink and Xilinx System Generator to simplify hardware implementation without extensive HDL coding. The system incorporated filtering, noise reduction, and signal enhancement techniques to improve ECG signal quality. Their implementation showed improved performance compared to software-based methods; however, it mainly focused on preprocessing rather than complete abnormality detection. [2]

Muhammed Yildirim and Ahmet Çinar examined real-time signal processing techniques implemented on FPGA hardware. Their work demonstrated the advantages of parallel processing by implementing filtering, transformation, and threshold-based operations using Verilog and Vivado tools. The system showed significant improvements in processing speed compared to traditional CPU-based methods, highlighting the suitability of FPGA for time-critical biomedical applications. [3]

J.S. Swamy and S. Sakkara introduced an FPGA-based biomedical signal processing system for disease detection applications. Their approach emphasized improving signal clarity through enhancement and noise suppression techniques. The system demonstrated that FPGA platforms can efficiently process complex biological signals with reduced delay and power consumption. However, the design relied on conventional processing techniques and did not incorporate neuromorphic computing models. [4]

Galvis-Chacón et al. proposed an ECG classification system using Spiking Neural Networks, where temporal spike-based processing was used to capture dynamic patterns in cardiac signals. Their approach demonstrated improved efficiency and reduced computational complexity compared to conventional neural networks. However, the implementation was primarily software-based and lacked complete hardware realization on FPGA platforms. [5]

S. Roy et al. introduced a neuromorphic computing approach for biomedical signal processing using spiking neural networks. Their system utilized event-driven computation to process ECG signals efficiently, reducing power consumption and latency. The results showed that neuromorphic models are highly suitable for real-time biomedical applications; however, the system lacked full integration with FPGA-based deployment. [6]

U.R. Nela Kuditi, M.N. Babu, and T.N. Bhagirath presented an FPGA-based co-simulation environment for real-time signal processing. Their work focused on optimizing hardware resource utilization while maintaining high throughput. The results demonstrated that FPGA implementations significantly outperform conventional software approaches in terms of speed and efficiency, making them suitable for real-time biomedical applications. [7]

Anirudh Kumar et al. proposed a hardware-efficient ECG classification system using deep learning techniques implemented on FPGA. Their system focused on optimizing resource utilization while maintaining high classification accuracy. The design demonstrated improved performance in real-time processing compared to traditional CPU-based implementations. However, the complexity of deep learning models increased hardware resource consumption. [8]

Another study proposed an FPGA-based implementation of Spiking Neural Networks for real-time biomedical signal classification. The system demonstrated efficient parallel processing and reduced latency, highlighting the advantages of neuromorphic computing for embedded healthcare applications. However, scalability and adaptive learning mechanisms were limited in the implementation. [9]

While these studies demonstrate the effectiveness of FPGA-based systems, deep learning techniques, and neuromorphic approaches, most of them are limited to either signal preprocessing, computationally intensive models, or partial hardware implementations. In contrast, the proposed work focuses on a complete neuromorphic ECG abnormality detection system by integrating a Spiking Neural Network based on LIF neurons with FPGA hardware. The system utilizes AXI DMA communication and PS-PL co-design on the PYNQ-Z2 platform, enabling real-time processing, reduced latency, and efficient hardware utilization for biomedical applications.

2.2 NOVELTY OF THE PRESENT WORK

Neuromorphic ECG Processing Architecture: Unlike conventional ECG analysis systems that rely on continuous-valued computations, this project introduces a neuromorphic architecture based on Spiking Neural Networks (SNNs) using the Leaky Integrate-and-Fire (LIF) neuron model. The system processes ECG signals in the form of spikes, closely mimicking biological neural behavior. This approach enables efficient temporal signal analysis and reduces computational complexity compared to traditional neural network models.

Hardware-Software Co-Design (PS-PL Integration): The proposed system implements a complete PS-PL co-design architecture on the PYNQ-Z2 platform, where the Jupyter Notebook interface running on the ARM Cortex-A9 processor handles data loading and visualization, while the Programmable Logic performs neuromorphic computation. High-speed data transfer is achieved using AXI Direct Memory Access (DMA), enabling efficient communication between software and hardware without heavy processor involvement.

ECG Abnormality Detection Using Sample-by-Sample Processing: The system is designed to process ECG samples continuously through the FPGA-based SNN in a sampleby-sample manner. Due to the inherent parallelism of FPGA architecture, multiple neurons operate simultaneously, significantly reducing latency. The system generates spike counts and detects abnormalities based on predefined thresholds, making it suitable for continuous cardiac monitoring applications.

Efficient Hardware Implementation of SNN: The neuromorphic model is implemented using Verilog RTL design, including modules such as LIF neurons, weight ROM, and a top-level integration block. Unlike many existing works that remain in software simulation, this project demonstrates a complete hardware realization of SNN, ensuring efficient resource utilization, low power consumption, and scalability for embedded systems.

Integration with Real ECG Dataset: The proposed system is validated using real ECG signals from the MIT-BIH Arrhythmia Database, ensuring practical applicability. The ECG samples are normalized and processed through the hardware accelerator using a Jupyter Notebook interface on the PYNQ-Z2 platform, where results such as waveform visualization and abnormality detection are displayed. This end-to-end
Pipeline from dataset to hardware output makes the system more robust and application oriented.

Lightweight and Energy-Efficient Design: Compared to deep learning-based ECG classification systems, the proposed neuromorphic approach reduces computational overhead and memory requirements. The use of spike-based processing and simple neuron models ensures lower power consumption, making the system suitable for portable and wearable healthcare devices. The system achieves its classification objective using only 10 LIF neurons with predefined weights, demonstrating that effective ECG abnormality detection can be accomplished with minimal hardware resources.

By incorporating these features, the proposed work presents a complete neuromorphic ECG abnormality detection system on FPGA, combining SNN-based processing, hardware-accelerated performance, and hardware efficiency. It addresses the limitations of traditional software based and deep learning approaches by providing a scalable, low latency, and energy efficient solution for biomedical signal processing applications.

CHAPTER 3

ANALYSIS AND DESIGN

The design and development of the neuromorphic ECG abnormality detection system follow a hardware-software co-design approach that integrates Verilog RTL implementation with the PYNQ-Z2 platform for FPGA-based SNN deployment. This chapter outlines the key stages in system analysis and the corresponding design methodology in detail.

System Requirements Analysis: The initial phase focuses on gathering comprehensive system requirements relevant to cardiac monitoring applications. The system must process ECG signal samples from the MIT-BIH Arrhythmia Database and classify each sample as normal or abnormal using a neuromorphic Spiking Neural Network. The neuromorphic hardware must implement biologically inspired LIF neuron dynamics including membrane potential integration, threshold detection, spike generation, and membrane leakage. Additional requirements such as power efficiency, FPGA resource utilization, and scalability for embedded deployment are also considered during the analysis phase.

Algorithm Selection and Neuron Modeling: The Leaky Integrate-and-Fire neuron model is selected as the core computational unit due to its simplicity, biological plausibility, and suitability for hardware implementation. Each LIF neuron accumulates weighted input from the ECG sample and generates a spike when the membrane potential exceeds a predefined threshold of 800. A leak factor of 20 is applied at each clock cycle to model the natural decay of membrane potential. Ten such neurons are instantiated in the SNN architecture, each with unique synaptic weights stored in a weight ROM module. The aggregate spike count across all neurons is used to determine the abnormality status of each ECG sample.

RTL Design and Modular Integration: With the algorithm defined, the neuromorphic system is implemented using Verilog RTL design. The system is structured modularly, with each functional unit encapsulated as an independent module to support design reuse and easier debugging. The lif_neuron.v module implements the core neuron dynamics, the weight_rom.v module stores the synaptic weights, and the snn_top.v module integrates all components with an AXI Stream interface for communication with the AXI DMA. The Vivado block design connects the Zynq PS, AXI DMA, and SNN RTL module through AXI Interconnect and AXI SmartConnect, enabling efficient PS-PL data transfer.

FPGA Synthesis and Implementation: After successful RTL design and block design validation in Vivado 2018.3, the system is synthesized and the bitstream is generated for deployment on the PYNQ-Z2 platform. The bitstream file and hardware handoff file are uploaded to the PYNQ Jupyter Notebook environment, where the PYNQ overlay loads the bitstream onto the FPGA fabric. The AXI DMA is accessed through PYNQ Python library, enabling sample-by-sample transfer of ECG data from the ARM processor to the neuromorphic hardware accelerator and back.

Testing, Validation and Results: The hardware is tested using 1000 ECG samples from MIT-BIH Arrhythmia Database Record 100, sampled at 360 Hz. The samples are normalized to a range of 0 to 1000 before being transferred to the SNN hardware through AXI DMA. The spike count and abnormality flag returned by the hardware are recorded for each sample and visualized through a Python-based ECG waveform plot, where normal samples are displayed in blue and abnormal samples are highlighted in red. The testing phase validates the end-to-end pipeline from dataset loading to hardware output, confirming the functional correctness and practical applicability of the proposed neuromorphic ECG abnormality detection system.

3.1 NEUROMORPHIC SYSTEM

The neuromorphic system forms the core of the proposed design. It is based on the concept of Spiking Neural Networks (SNNs), which are inspired by biological neural systems. In this approach, ECG signals are converted into spike-based representations and processed using neuron models that closely mimic the behavior of biological neurons in the human brain.

SNNs differ from traditional neural networks in that they operate using discrete spike events rather than continuous values. In conventional artificial neural networks (ANNs), neurons compute weighted sums of inputs and apply activation functions such as ReLU or sigmoid to produce continuous-valued outputs. In contrast, SNN neurons accumulate input over time and generate binary spike events only when their internal state (membrane potential) exceeds a threshold. This fundamental difference makes SNNs highly efficient for hardware implementation and particularly suitable for time-based signal processing such as ECG analysis.

The event-driven nature of SNNs provides several advantages over traditional neural networks for biomedical applications. First, computation occurs only when spike events are generated, resulting in significantly reduced power consumption during periods of low neural activity. Second, the temporal dynamics of spike generation naturally encode time varying signals like ECG waveforms, enabling the network to capture temporal patterns without explicit feature extraction. Third, the binary nature of spike communication simplifies hardware implementation, as spike events can be represented using single-bit signals rather than multi-bit floating-point values.

The proposed neuromorphic system consists of three primary components: (1) the spike encoding mechanism that converts normalized ECG samples into weighted inputs for the neuron array; (2) the LIF neuron array that processes the weighted inputs and generates spike outputs based on membrane potential dynamics; and (3) the spike aggregation and classification logic that counts the total spikes across all neurons and determines the abnormality status of each ECG sample.

3.1.1 LIF Neuron Design

The Leaky Integrate-and-Fire (LIF) neuron is used as the fundamental processing unit in the system. It accumulates incoming ECG signal values and applies a leakage factor to simulate the natural decay of membrane potential observed in biological neurons. The LIF model strikes an optimal balance between biological plausibility and computational simplicity, making it the most widely adopted neuron model for hardware neuromorphic implementations.

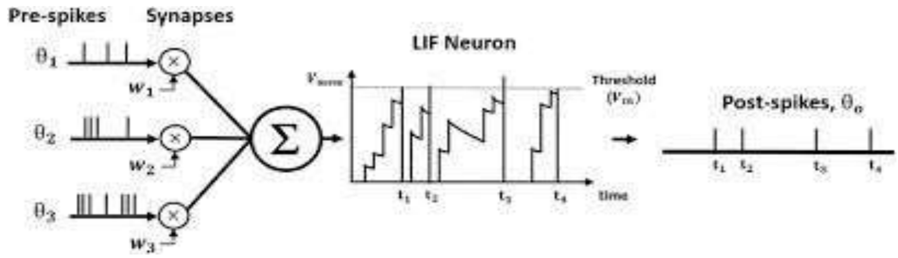

Figure 3.1 LIF neuron model showing weighted spike inputs, membrane potential integration with leakage, threshold crossing, and output spike generation.

The mathematical model of the LIF neuron is described by the following discrete-time equation:

V(t) = V(t-1) + W * Input - Leak

Where V(t) represents the membrane potential at time step t, W is the synaptic weight associated with the neuron, Input is the normalized ECG sample value, and Leak is a constant representing the natural decay of membrane potential. When the membrane potential V(t) exceeds a predefined threshold V_th, the neuron generates a spike output, and the membrane potential is reset to zero. This reset mechanism prevents the neuron from continuously firing and ensures that each spike event carries meaningful information about the signal dynamics input.

The leakage mechanism is a critical component of the LIF model that distinguishes it from simpler integrate-and-fire models. By continuously subtracting a small leak value from the membrane potential at each time step, the model ensures that the neuron does not accumulate charge indefinitely. This behavior mimics the ion channel leakage observed in biological neurons, where the membrane potential gradually returns to its resting state in the absence of input stimulation. The leak factor also introduces a temporal sensitivity to the neuron, as inputs that arrive in rapid succession are more likely to trigger a spike than inputs that are spread over longer time intervals.

Parameter	Value	Description
Threshold (V_th)	800	Membrane potential threshold for spike generation
Leakage Factor	20	Constant subtracted from membrane potential per clock cycle
Number of Neurons	10	Total LIF neurons in the SNN array
Data Width	16-bit	Bit width of membrane potential and input signals
Reset Value	0	Membrane potential value after spike generation
Weight Width	16-bit	Bit width of synaptic weight values

Table 3.1: LIF Neuron Parameters

The LIF neuron is implemented in Verilog RTL as a synchronous digital circuit. The module accepts a clock signal, an active-low reset signal, a 16-bit ECG input value, and produces a single-bit spike output along with the current membrane potential value. On each rising edge of the clock, the neuron updates its membrane potential by adding the weighted input and subtracting the leak factor. If the updated potential exceeds the threshold, the spike output is asserted and the potential is reset. The Verilog implementation ensures that all operations are performed using fixed-point arithmetic, avoiding the complexity and resource overhead of floating-point computation.

The choice of parameter values is guided by both the characteristics of the ECG signal and the constraints of the hardware implementation. The threshold value of 800 is selected to ensure that the neuron fires only when the input signal contains sufficient energy, thereby filtering out low-amplitude noise. The leak factor of 20 provides a moderate decay rate that allows the neuron to respond to sustained input patterns while preventing false spikes from transient noise. The 16-bit data width provides sufficient dynamic range to represent the normalized ECG sample values (0 to 1000) and the membrane potential without overflow.

3.1.2 Spiking Principle

The spiking mechanism is the fundamental operation that distinguishes neuromorphic computing from conventional computing paradigms. In the proposed system, the spiking principle is based on threshold comparison: when the membrane potential of a LIF neuron crosses the predefined threshold, a spike event is generated. Otherwise, the neuron continues integrating inputs and the membrane potential continues to evolve according to the LIF dynamics.

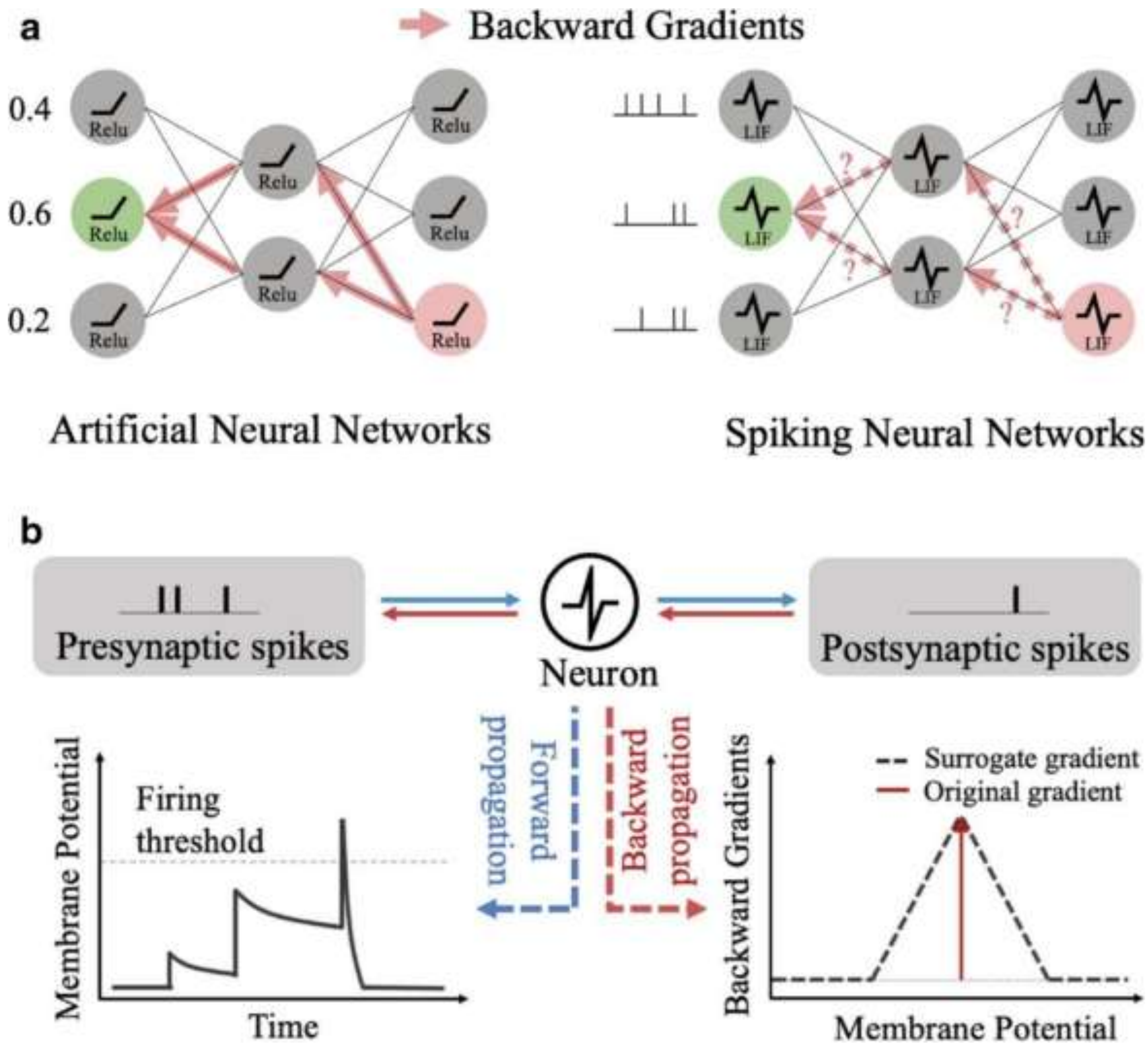

Fig 3.2 ANN vs SNN gradient propagation with LIF neuron behavior and surrogate gradient learning.

Spike-based processing offers several key advantages for ECG signal analysis:

- **Efficient Temporal Representation:** The timing and frequency of spikes naturally encode the temporal dynamics of the ECG signal. Higher-amplitude ECG samples cause faster membrane potential accumulation, resulting in more frequent spike generation. This temporal coding mechanism allows the SNN to distinguish between normal and abnormal ECG patterns based on spike activity patterns.

- **Reduced Computation:** Unlike conventional neural networks that perform multiply-accumulate operations on every input, SNN neurons only generate output (spikes) when the threshold is exceeded. This event-driven computation significantly reduces the total number of operations, leading to lower power consumption and improved energy efficiency.

- **Hardware-Friendly Implementation:** Spike events are binary signals that can be represented using single-bit values, simplifying the hardware implementation. The threshold comparison and reset operations are straightforward digital logic operations that can be efficiently implemented on FPGA platforms using minimal hardware resources.
- **Natural Noise Filtering:** The threshold mechanism inherently provides a degree of noise filtering, as low-amplitude noise signals are unlikely to cause the membrane potential to exceed the threshold. This built-in noise immunity reduces the need for complex preprocessing stages and simplifies the overall system design. The spike count is used as the primary metric for abnormality detection in the proposed system. For each ECG sample, the total number of spikes generated across all 10 neurons is computed. If the aggregate spike count exceeds a predefined abnormality threshold of 2, the sample is classified as abnormal. Otherwise, it is classified as normal. This threshold based classification approach is simple yet effective, as it leverages the collective response of the neuron array to distinguish between normal and abnormal ECG patterns.

The relationship between spike count and ECG signal characteristics can be understood as follows: normal ECG samples typically have moderate amplitude values that produce a controlled number of spikes across the neuron array. In contrast, abnormal ECG samples often exhibit higher amplitude values, irregular patterns, or sudden changes that cause increased neural activity, resulting in higher spike counts. By monitoring the aggregate spike count, the system can effectively identify samples that deviate from normal cardiac behavior.

3.2 FPGA IMPLEMENTATION

The FPGA implementation integrates neuromorphic processing with hardware modules to achieve high-speed signal processing. Programmable logic performs parallel computation, improving efficiency compared to software-based systems. The implementation leverages the reconfigurability and parallelism of the FPGA fabric to realize a complete neuromorphic processing pipeline that operates in real time.

3.2.1 RTL Design of SNN

The neuromorphic system is implemented using Register Transfer Level (RTL) design in Verilog. The design is modular and consists of multiple components that work together to process ECG signals and generate spike-based outputs. The modular architecture facilitates independent testing and verification of each component, simplifies debugging, and enables design reuse for future extensions.

The primary modules in the RTL design include: **LIF Neuron Module (lif_neuron.v):** This module implements the Leaky Integrate and-Fire neuron model. It receives ECG input values, updates the membrane potential by adding the weighted input and subtracting the leak factor, and generates a spike when the threshold is exceeded. The neuron resets its potential after firing, ensuring proper operation for the next input sample. The module is designed as a reusable component that can be instantiated multiple times to create the neuron array.

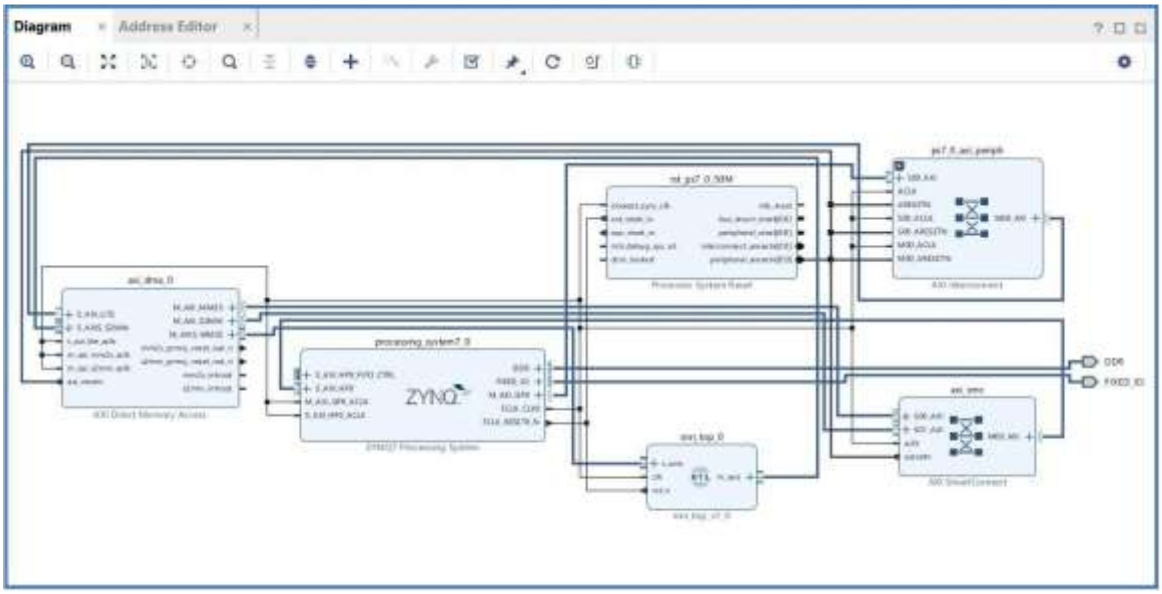

Fig3.3 ZYNQ-based Vivado block design with AXI DMA and custom IP integration.

Weight ROM Module (weight_rom.v): This module stores predefined synaptic weights used in neuron computation. The weights are organized as a read-only memory (ROM) with 10 entries, one for each neuron in the array. Each weight is a 16-bit value that determines the contribution of the ECG signal to the corresponding neuron membrane potential. The weight values are initialized during synthesis and remain constant during operation. In future implementations, these weights can be made programmable to support on-chip learning.

Top Module (snn_top.v): The top module integrates multiple LIF neurons and manages the overall data flow. It includes AXI Stream interfaces for communication with the DMA module, enabling seamless integration with the Zynq PS-PL co-design architecture. The module processes input ECG samples by distributing weighted versions of the input to each neuron, collecting spike outputs, computing the aggregate spike count, and determining the abnormality status. The output is packed into a 32-bit format for transmission through the AXI Stream interface.

The output format of the system is defined as:

m_axis_tdata[4:0] = {abnormality, spike_count}

where the most significant bit (bit 4) represents the abnormality detection flag and the remaining four bits (bits 3:0) represent the spike count. The remaining bits of the 32-bit output word are reserved for future use.

The RTL design is synthesized and implemented using Vivado Design Suite version 2018.3. The synthesis process translates the Verilog RTL description into a gate-level netlist, which is then mapped onto the FPGA fabric through the implementation process. The modular structure of the design ensures efficient utilization of FPGA resources including lookup tables (LUTs), flip-flops, and block RAM.

3.2.2 AXI DMA Communication

AXI Direct Memory Access (DMA) is used for high-speed data transfer between the Processing System and Programmable Logic. It enables efficient communication without heavily involving the processor, thereby improving system performance. The AXI DMA operates as an intermediary between the PS memory space and the PL data processing pipeline, facilitating seamless transfer of ECG samples to the neuromorphic hardware and retrieval of processed results.

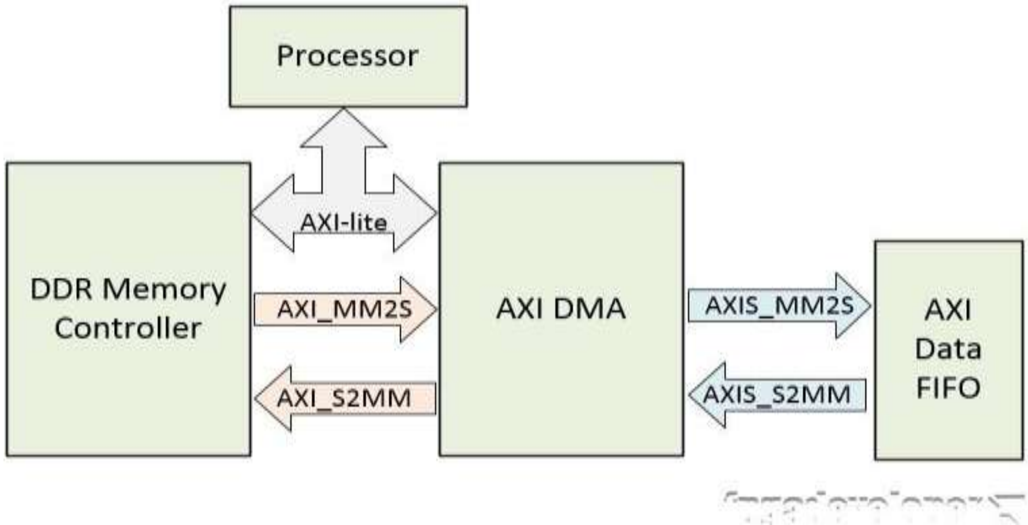

Figure 3.4: The RTL block design for a hardware system on a Zynq-7020 platform, showing the data flow between a processor, DDR memory controller, and an AXI DMA engine.

In this project, the AXI DMA operates in simple mode with a 32-bit data width. The DMA engine supports two primary data transfer channels:

- **Memory-Mapped to Stream (MM2S):** This channel transfers data from the PS memory to the PL through the AXI Stream interface. ECG samples stored in the PS memory are read by the DMA engine and streamed to the snn_top module for neuromorphic processing.
- **Stream to Memory-Mapped (S2MM):** This channel transfers processed data from the PL back to the PS memory. The spike count and abnormality flag generated by the snn_top module are streamed through this channel and stored in the PS memory for visualization and analysis.

The communication process involves the following sequential steps:

- ECG data is loaded in the Processing System using Python through PYNQ Jupyter Notebook interface.
- The normalized ECG samples are written to a contiguous memory buffer allocated using thePYNQ allocate function.
- The AXI DMA MM2S channel is configured to transfer the ECG data from the memory buffer to the snn_top module.

- The neuromorphic system processes each sample through the 10-neuron LIF array and generates spike count and abnormal outputs.
- The AXI DMA S2MM channel transfers the processed output back to a separate memory buffer

in the PS.

- The results are read from the output buffer and visualized using matplotlib plots in the Jupyter Notebook.

The use of AXI DMA ensures fast and efficient data transfer, reduces processor workload, and supports continuous streaming of data. The DMA operates independently of the ARM processor once configured, allowing the processor to perform other tasks while data transfer is in progress. This non-blocking operation is essential for achieving real-time performance in the proposed system.

3.3 SYSTEM ARCHITECTURE

The system architecture of the proposed neuromorphic ECG abnormality detection system defines the overall organization of the system and the flow of data between hardware and software components. The system is designed using a hardware-software co-design approach, where the Processing System (PS) handles control and data management, while the Programmable Logic (PL) performs neuromorphic computation.

The Zynq-7020 SoC architecture provides the foundation for system design. The PS side consists of a dual-core ARM Cortex-A9 processor running at up to 667 MHz, with associated peripherals including DDR3 memory controller, I/O peripherals, and interrupt controller. The PL side consists of Artix-7 equivalent FPGA fabric with 53,200 lookup tables (LUTs), 106,400 flip-flops, 140 block RAM tiles (36 Kb each), and 220 DSP slices. The PS and PL are connected through multiple high-performance AXI interfaces that enable efficient data transfer between the processor and the reconfigurable logic.

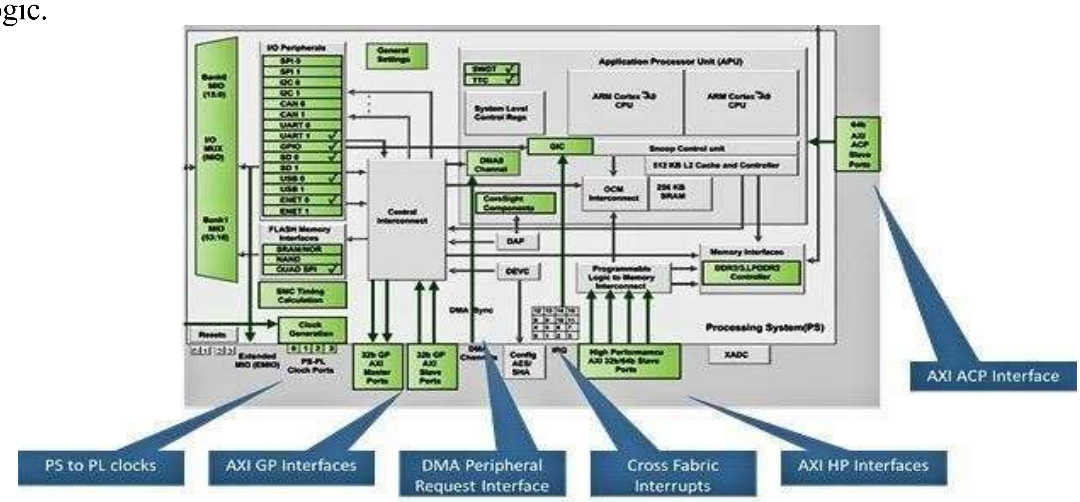

Figure 3.5: The data flow in an AXI DMA communication architecture between processor, memory and peripherals

The system consists of several key components:

- **ECG Data Input:** ECG signals from the MIT-BIH Arrhythmia Database are loaded into the PS memory through the Jupyter Notebook interface. The signals are normalized to a range of 0 to 1000 to match the input requirements of the neuromorphic hardware.
- **Processing System (PS):** The ARM Cortex-A9 processor runs the PYNQ Linux operating system and hosts the Jupyter Notebook server. It manages data loading, memory allocation, DMA configuration, and result visualization.
- **AXI DMA:** Facilitates high-speed data transfer between the PS memory and the PL neuromorphic processing unit without significant processor involvement.
- **Neuromorphic Processing Unit (PL):** The custom SNN hardware implemented in the FPGA programmable logic. It consists of the weight ROM, 10 LIF neurons, spike counting logic, and abnormality detection logic.
- **Output Visualization:** The processed results (spike counts and abnormality flags) are transferred back to the PS and visualized using matplotlib plots in the Jupyter Notebook, showing normal (blue) and abnormal (red) ECG segments.

The data flow of the system is as follows: ECG data is loaded and normalized in the PS; the normalized samples are transferred to the PL through AXI DMA; the neuromorphic system processes each sample using the 10-neuron LIF array; spike counts are generated for each sample; abnormality is detected based on the spike count threshold; the results are transferred back to the PS; and finally, the results are visualized using graphical plots. This end-to-end pipeline demonstrates the complete integration of data acquisition, neuromorphic processing, and result visualization on a single hardware platform.

3.4 PYNQ-Z2 DEVELOPMENT BOARD

The PYNQ-Z2 development board is based on the Xilinx Zynq-7020 System-on-Chip (SoC), which integrates a Processing System (PS) and Programmable Logic (PL). It provides a wide range of interfaces that enable communication with external devices and support efficient system implementation. The PYNQ framework provides a Python-based programming environment that simplifies the interaction between software applications and FPGA hardware, making it an ideal platform for rapid prototyping and educational purposes.

The key interfaces and components of the PYNQ-Z2 board are described below:

Ethernet Port: The Ethernet port provides network connectivity between the board and a host computer. In this project, it is used to access the Jupyter Notebook interface through a web browser by configuring the board IP address. It plays a key role in enabling communication, data transfer, and system control.

USB Ports: The board includes USB ports that allow connection of peripheral devices such as keyboards, mice, and storage devices. These ports are managed by the Processing

System and provide flexibility for
system interaction.

HDMI Ports: The HDMI input and output ports support high-definition video processing. Although not directly used in this project, they can be utilized for real-time display of ECG waveforms and detection results in future extensions.

Pmod Connectors: Pmod connectors are general-purpose interfaces used for connecting external modules such as sensors and communication devices. They support protocols like GPIO, SPI, I2C, and UART, making them highly flexible for system expansion and sensor integration.

DDR Memory: The board includes 512 MB of DDR3 memory, which is used for storing data and executing applications. It is essential for handling large datasets such as ECG signals and for buffering data during DMA transfers.

AXI Interfaces: AXI interfaces provide high-speed communication between the Processing System and Programmable Logic. In this project, AXI DMA is used to transfer ECG data efficiently between software and hardware,
ensuring fast processing.

LEDs, Switches, and Buttons: The onboard LEDs, slide switches, and push buttons are used for debugging and user interaction. LEDs indicate system status, while switches and buttons provide manual control inputs for testing and verification.

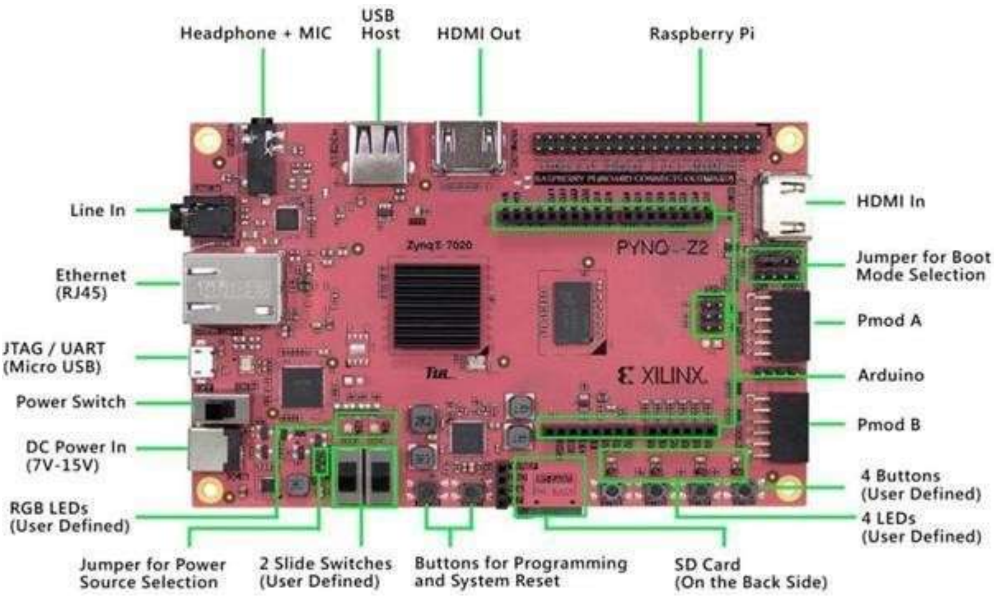

Figure 3.6: PYNQ-Z2 Development Board

CHAPTER 4

IMPLEMENTATION

The implementation of the proposed neuromorphic ECG abnormality detection system is carried out using a hardware–software co-design approach on the PYNQ-Z2 FPGA platform. The system combines FPGA-based neuromorphic computation with software-based control and visualization to achieve efficient and scalable ECG signal processing.

The ECG signals used in this system are obtained from the MIT-BIH Arrhythmia Database, which contains real-time cardiac recordings. These signals are processed using a Spiking Neural Network (SNN) implemented in hardware using Verilog RTL. The implementation involves multiple stages including data acquisition, preprocessing, RTL design, hardware integration, and execution using the PYNQ framework.

The main goal of this implementation is to demonstrate how neuromorphic computing principles can be efficiently mapped onto FPGA hardware to achieve parallel processing and reduced computational complexity compared to traditional software-based approaches.

4.1 System Implementation Overview

The overall system is designed using a hardware–software co-design architecture, where the Processing System (PS) and Programmable Logic (PL) work together to perform ECG abnormality detection.

The Processing System, which consists of an ARM Cortex-A9 processor, is responsible for handling high-level operations such as loading ECG data, preprocessing, managing data transfer, and visualizing results. The Programmable Logic implements the Spiking Neural Network, which neuromorphic computation performs the core

The ECG data is first loaded into the system and normalized to match the input requirements of the hardware. The normalized data is then transferred to the FPGA using AXI DMA, which ensures high-speed communication between the PS and PL.

Inside the FPGA, the SNN processes the ECG signal using parallel neuron computation. The output results, including spike count and abnormality detection, are transferred back to the PS for further analysis.

This approach enables efficient utilization of hardware resources while maintaining flexibility through software control.

Fig 4.1: Overall System Architecture of Neuromorphic ECG Abnormality Detection System

4.2 RTL Design and Module Implementation

The neuromorphic processing system is implemented using Verilog RTL design. The system is composed of multiple modules that together form a Spiking Neural Network capable of processing ECG signals.

The fundamental unit of the system is the Leaky Integrate-and-Fire (LIF) neuron. This neuron integrates incoming signals over time and generates a spike when the membrane potential exceeds a predefined threshold. A leakage parameter is included to model the natural decay of the membrane potential, preventing continuous accumulation.

In this implementation, a total of 10 LIF neurons are used. Each neuron processes the input signal independently, enabling parallel computation. This parallelism significantly improves processing efficiency and reduces latency.

The input ECG signal is scaled using predefined weights stored in a ROM module. These weights ensure that each neuron receives a slightly different version of the input signal, allowing the system to capture variations in ECG patterns.

The top module integrates all functional components, including input interface, neuron array, weight scaling, spike generation, and output logic. It ensures synchronization and proper data flow across all modules.

The key design parameters used in the system include:

- Threshold value (V_THRESH = 800)
- Leakage factor (LEAK = 20)
- Number of neurons = 10

These parameters are selected based on experimental observation to ensure stable spike generation and effective classification

4.3 Hardware Integration Using Vivado

The RTL modules are integrated into a complete hardware system using the Vivado Design Suite. The system is designed using block design methodology, which allows easy integration of different components.

The Zynq Processing System is configured to communicate with the Programmable Logic using AXI interfaces. The AXI DMA module is used for high-speed data transfer between the PS and PL. It enables efficient streaming of ECG data without excessive processor involvement.

AXI interconnect and SmartConnect modules are used to establish communication between various components in the system. The custom SNN module is connected to the AXI DMA using AXI Stream interface, ensuring continuous data flow.

The system is configured in simple DMA mode with 32-bit data width. The High-Performance (HP) port of the Zynq PS is enabled to support faster data transfer.

After integration, synthesis and implementation are performed, and a bitstream is generated. This bitstream is used to configure the FPGA and execute the hardware design

4.4 Data Flow and Processing Mechanism

The system follows a structured data flow pipeline to ensure efficient processing of ECG signals.

Initially, ECG samples are loaded into the Processing System using Python libraries. The signals are normalized to a range suitable for hardware processing.

The normalized data is transferred to the FPGA using AXI DMA. The DMA controller sends the data in a streaming manner to the Programmable Logic.

Inside the FPGA, the input signal is distributed across the neuron array. Each neuron processes the input independently and generates spike outputs based on its threshold and leakage characteristics.

The outputs of all neurons are combined to compute the total spike count. This value represents the neural activity of the system.

A threshold-based classification mechanism is applied:

- Spike count > threshold → Abnormal
- Spike count ≤ threshold → Normal

The result is packed into a structured format and sent back to the Processing System through AXI DMA.

4.5 Execution on PYNQ Platform

The execution of the system is performed using the PYNQ framework, which provides a Python-based interface for interacting with FPGA hardware.

The bitstream generated from Vivado is loaded onto the FPGA using the Overlay class. This step configures the hardware and makes the SNN ready for execution.

The AXI DMA module is initialized in the Jupyter Notebook environment to facilitate communication between the Processing System and Programmable Logic. Memory buffers are allocated for sending and receiving data.

The ECG dataset is loaded using the WFDB library. The signals are normalized and processed sample-by-sample through the FPGA.

Each ECG sample is transferred to the FPGA using DMA, where it is processed by the Spiking Neural Network. The output data, including spike count and abnormality detection, is received back in the Processing System.

Although visualization is performed in Jupyter Notebook, the output graphs are presented in the Results chapter.

This execution process demonstrates seamless interaction between hardware and software components.

4.6 System Workflow and Operation

The overall operation of the proposed neuromorphic ECG abnormality detection system follows a structured workflow that integrates both hardware design and execution stages.

The process begins with the development of the neuromorphic architecture using Verilog RTL, where modules such as LIF neurons, weight ROM, and the top-level integration are designed and verified. These modules are then integrated using the Vivado Design Suite to form a complete hardware system.

After successful synthesis and implementation, the generated bitstream is deployed onto the PYNQ-Z2 FPGA board. The execution phase is controlled using the PYNQ framework through Jupyter Notebook, which enables interaction with the hardware.

During execution, ECG signals from the MIT-BIH dataset are loaded and preprocessed by normalization. The processed data is then transferred to the FPGA using AXI DMA, where it is handled in a streaming manner.

Inside the FPGA, the Spiking Neural Network processes the input signals using parallel LIF neurons. Each neuron generates spike outputs based on its internal threshold and leakage characteristics. The spike outputs are combined to compute the total spike count.

A classification decision is made based on the spike count. If the spike count exceeds the predefined threshold, the ECG signal is classified as abnormal; otherwise, it is considered normal.

The results are then transferred back to the Processing System and analyzed. Although visualization is performed in the execution stage, the graphical results are presented in the Results chapter.

This workflow demonstrates the complete operation of the system, highlighting the interaction between hardware and software components.

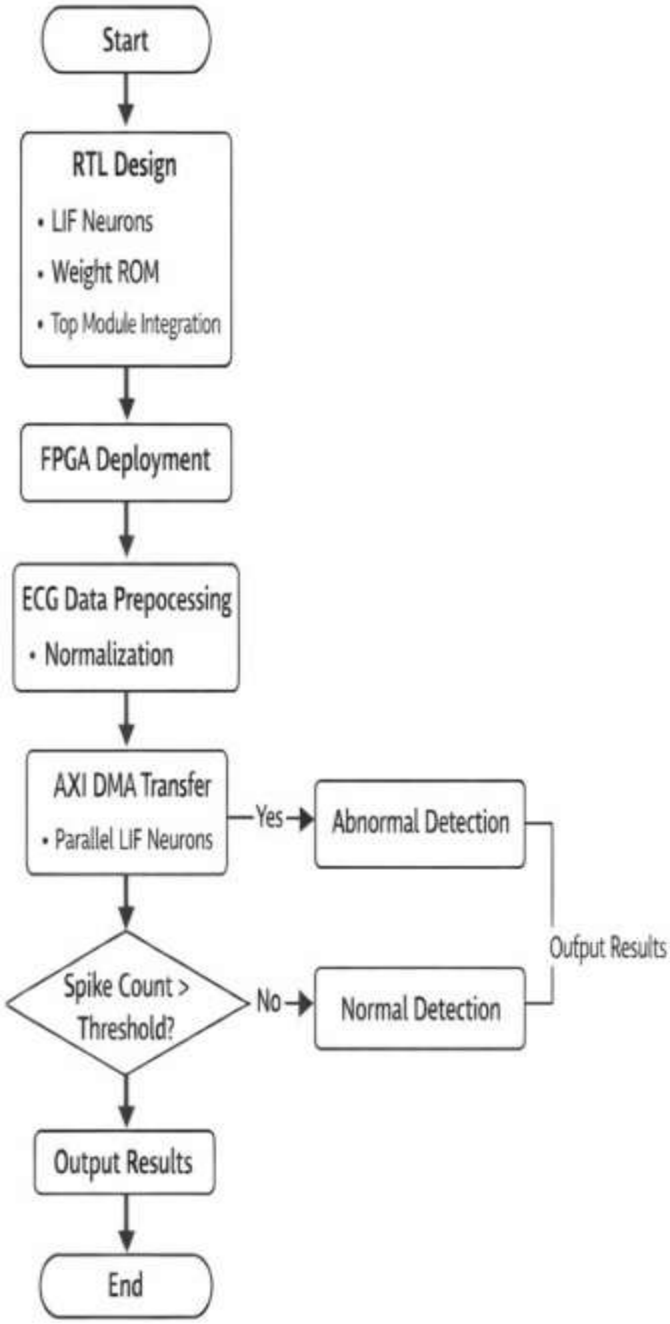

Fig 4.2 Flowchart of FPGA-based ECG classification using LIF neurons and spike threshold detection.

CHAPTER 5

RESULTS

This chapter presents the results obtained from the implementation of the neuromorphic ECG abnormality detection system on the PYNQ-Z2 FPGA platform. The system is evaluated based on simulation results, synthesis reports, and implementation performance. The ECG data used for testing is taken from the MIT-BIH Arrhythmia Database Record 100, and the system processes the input signals using a Spiking Neural Network to produce outputs in terms of spike count and abnormality detection.

5.1 SIMULATION RESULTS

The functional simulation of the proposed neuromorphic ECG abnormality detection system is performed using Vivado simulation tools to verify the correctness of the RTL design before hardware implementation. The simulation validates the behavior of all key signals, including the clock (clk), reset (rst_n), ECG input signal (ecg_input[15:0]), spike count (spike_count[3:0]), and abnormality output signal.

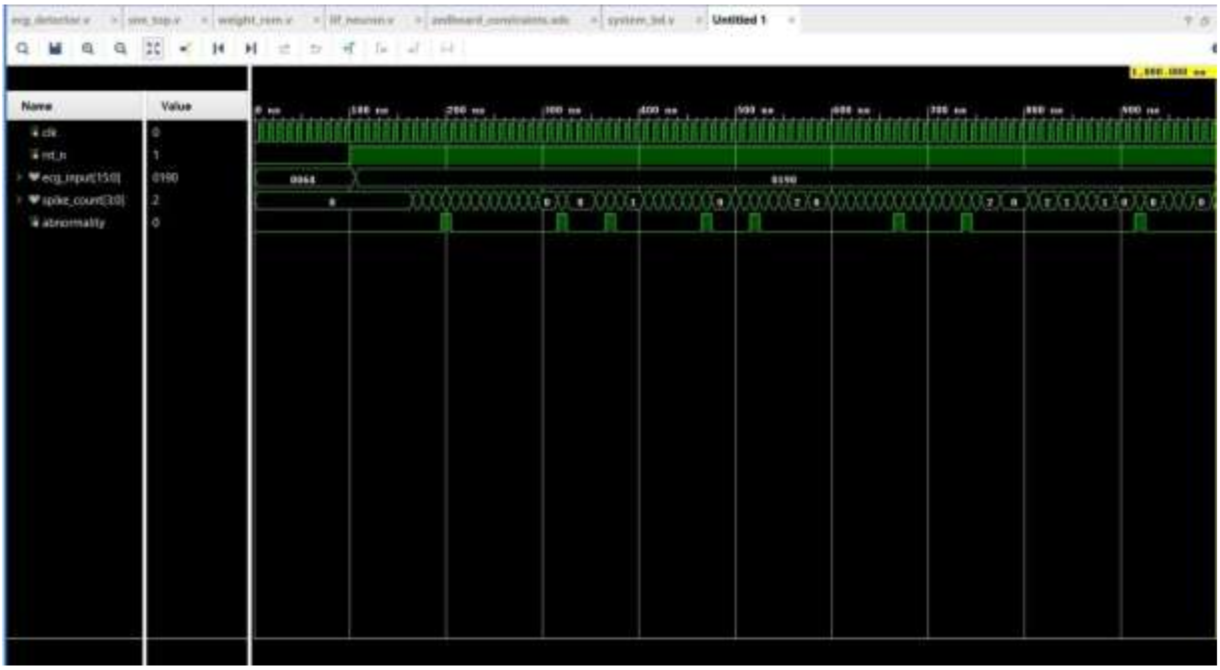

Fig 5.1 Vivado simulation waveform results

At the initial stage of simulation, the reset signal (rst_n) is asserted low, ensuring that all internal registers and outputs are initialized to known values. During this period, the spike count and abnormal outputs remain at zero, indicating proper system initialization. Once the reset signal is de-asserted (set to logic high), the system begins normal operation, and the ECG signal is applied.

The ECG input (ecg_input) is observed to change over time, representing different sample values of the ECG signal. These input values are processed by the neuromorphic system implemented using LIF neurons. Each neuron receives a weighted version of the input signal and updates its membrane potential accordingly. When the membrane potential exceeds the predefined threshold of 800, a spike is generated by the corresponding neuron.

The spike outputs from all 10 neurons are combined to produce the total spike count. The simulation waveform shows that the spike count varies dynamically based on the ECG signal input, reflecting the activity of the neuron array. For lower input values, fewer spikes are generated, resulting in a lower spike count. As the input signal increases in amplitude, neuron activity also increases, leading to a higher spiking count. This behavior confirms that the neuromorphic system correctly responds to variations in the input signal.

The abnormality detection logic is implemented using a threshold-based comparison. When the spike count exceeds the predefined threshold value of 2, the abnormality signal is asserted high (logic 1). This behavior is clearly observed in the simulation waveform, where the abnormal signal transitions to logic high during instances of increased spike activity corresponding to higher-amplitude ECG input values. When the spike count falls below or equals the threshold, the abnormality signal returns to logic low (logic 0), indicating normal conditions.

The timing relationship between the input signal, spike count, and abnormal output confirms the correct operation of the system. The design responds accurately to variations in input and produces corresponding outputs within a single clock cycle, demonstrating the low-latency processing capability of the FPGA implementation. The absence of timing violations or metastability issues in the simulation confirms that the design meets the timing requirements for reliable operation.

Overall, the simulation results validate the functional correctness of the neuromorphic ECG abnormality detection system. The system successfully performs spike generation, spike accumulation, and abnormality detection as intended. These results confirm that the RTL design is functionally correct and ready for synthesis and implementation on the FPGA hardware.

5.2 SYNTHESIS REPORT

The synthesis of the proposed neuromorphic ECG abnormality detection system is carried out using the Vivado Design Suite version 2018.3. The RTL design, including the LIF neuron module, weight ROM, and top module, is synthesized and mapped onto the target FPGA device (Zynq-7020, part number XC7Z020-1CLG400C). The synthesis process translates the behavioral Verilog description into a gate-level netlist optimized for the target device architecture.

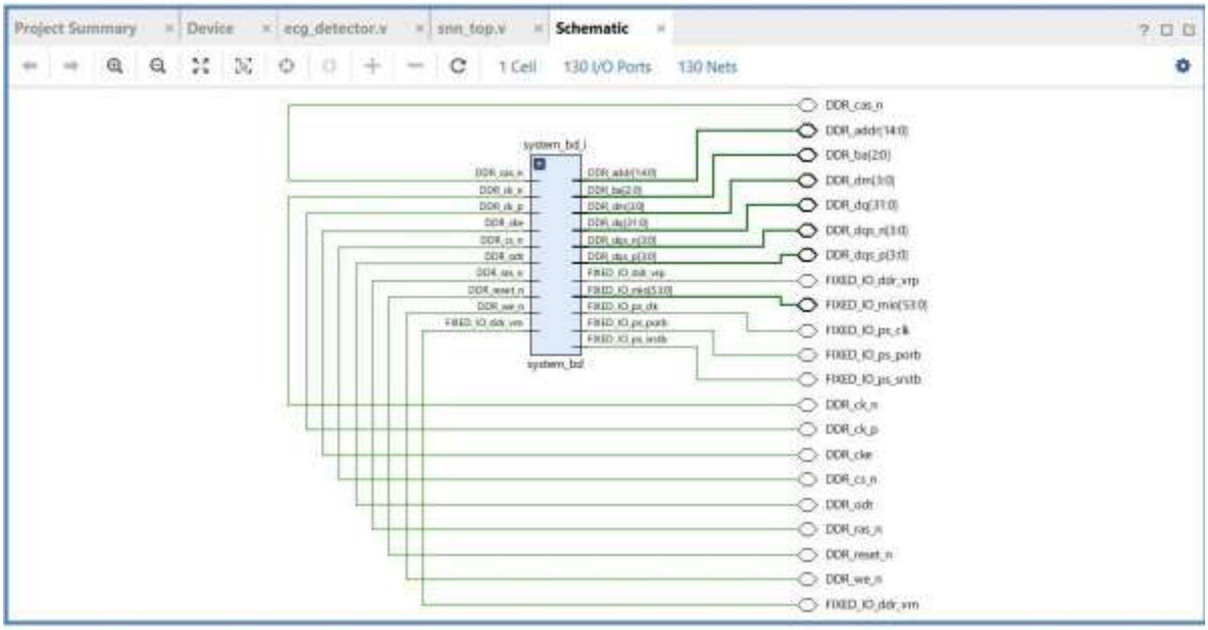

Fig5.2 Schematic diagram for an FPGA

The synthesis report confirms that all RTL modules are correctly mapped and interconnected without any errors or critical warnings. The design utilizes a small fraction of the available FPGA resources, demonstrating the lightweight nature of the neuromorphic approach. The resource utilization summary is presented below:

The DDR interface consists of multiple signals including address lines (DDR_addr), data lines

(DDR_dq), bank address (DDR_ba), data strobes (DDR_dqs), and control signals (DDR_ras, DDR_cas, DDR_we). These signals facilitate high-speed memory access required for data storage and processing during DMA operations.

The FIXED_IO interface represents the connection between the Processing System and external peripherals. It includes signals related to clock generation, system reset, and multiplexed I/O pins, enabling communication between the FPGA and the external environment. The synthesis process ensures that all these interfaces are correctly configured and connected.

The synthesis results indicate that the neuromorphic system requires minimal FPGA resources compared to the total available resources on the Zynq-7020 device. The 10neuron SNN with weight ROM and control logic occupies only a small percentage of the available LUTs and flip-flops, leaving substantial resources available for future extensions such as increasing the number of neurons, adding preprocessing stages, or implementing on-chip learning mechanisms.

The absence of errors in the synthesis report and the generated schematic confirms that the design is logically consistent, correctly interconnected, and ready for implementation. The synthesis results also demonstrate that the modular design approach facilitates efficient resource mapping and optimization by the Vivado synthesis engine.

5.3 IMPLEMENTATION PERFORMANCES

The implementation performance of the proposed system is evaluated by processing 1000 ECG samples from MIT-BIH Arrhythmia Database Record 100 through the FPGAbased neuromorphic hardware. The results are analyzed in terms of ECG waveform visualization, spike count analysis, and abnormality detection accuracy.

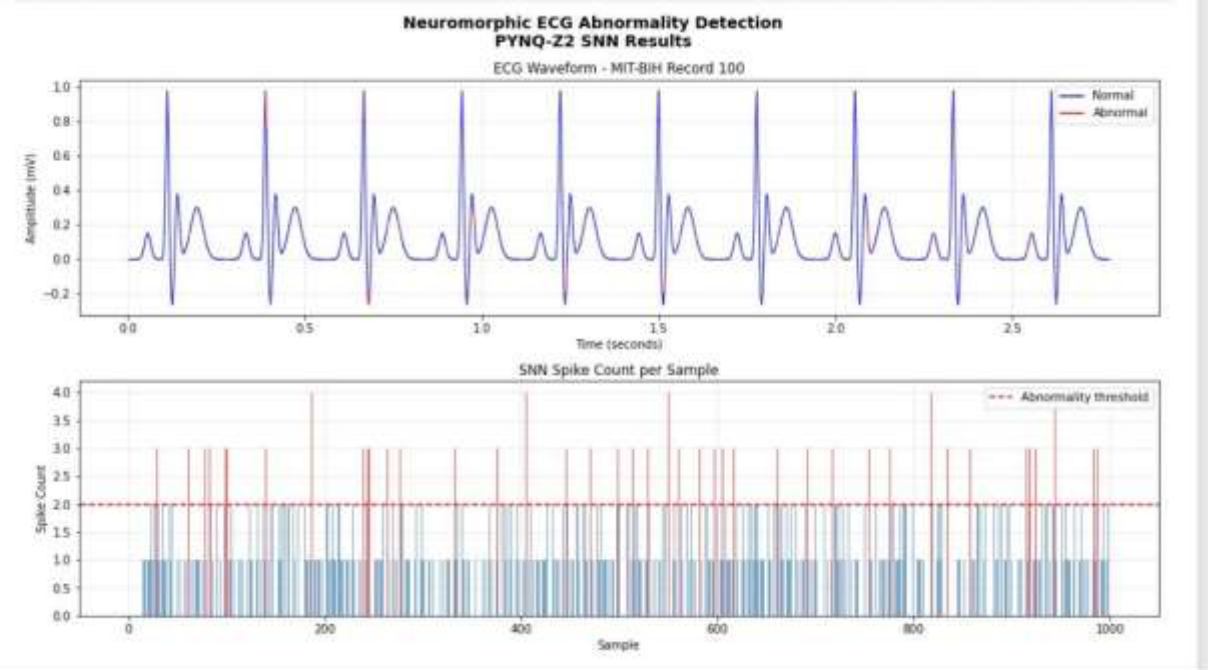

Fig 5.3 ECG waveform with abnormality detection results

The results demonstrate a strong correlation between spike activity and ECG signal behavior. Higher spike counts are associated with irregular or abnormal ECG patterns, while lower spike counts correspond to normal physiological signals. This confirms that the Spiking Neural Network effectively captures temporal variations in the ECG signal and translates them into meaningful spike-based representations.

The FPGA-based implementation ensures efficient processing by utilizing parallel computation of multiple neurons. All 10 LIF neurons process the input simultaneously within a single clock cycle, resulting in extremely low processing latency. The use of AXI DMA enables high-speed data transfer between the Processing System and Programmable Logic, reducing communication overhead and improving overall system throughput.

The system processes the ECG data sample-by-sample and generates real-time outputs, demonstrating the feasibility of neuromorphic computing for biomedical signal analysis. The simplicity of the threshold-based classification approach further enhances the efficiency of the system without compromising detection capability. The event-driven nature of the SNN ensures that computational resources are utilized only when spike events occur, contributing to the overall energy efficiency of the system.

The above table summarizes the key performance characteristics of the proposed neuromorphic ECG abnormality detection system. The system achieves efficient and reliable detection of abnormal ECG patterns using FPGA-based parallel processing with minimal hardware resources. The implementation demonstrates the effectiveness of Spiking Neural Networks in biomedical signal analysis while maintaining low computational complexity and efficient hardware utilization.

Parameter	**Observation**
Platform	PYNQ-Z2 (Zynq-7020 FPGA)
Dataset	MIT-BIH Arrhythmia Database (Record 100)
Number of Samples	1000 ECG samples
Sampling Rate	360 Hz
Processing Method	Spiking Neural Network (SNN)
Neuron Model	Leaky Integrate-and-Fire (LIF)
Number of Neurons	10
Threshold	Spike count > 2
Output Format	5-bit: {abnormality, spike_count[3:0]}
Detection Performance	Correct identification of abnormal regions
Processing Type	Parallel FPGA-based computation
Data Transfer	AXI DMA (32-bit, simple mode)
Processing Latency	Single clock cycle per sample
Power Efficiency	Low (event-driven spike computation)

Table 5.1: *Performance Summary of Proposed Neuromorphic ECG Detection System*

The proposed system shows strong potential for real-time healthcare applications with further optimization. The lightweight architecture, combined with the energy-efficient spike-based processing paradigm, makes the system suitable for deployment in portable and wearable cardiac monitoring devices where power consumption and processing latency are critical design constraints.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project successfully designed and implemented a neuromorphic ECG abnormality detection system on the PYNQ-Z2 FPGA platform using a Spiking Neural Network based on the Leaky Integrate-and-Fire neuron model. The system demonstrates the feasibility of neuromorphic computing for real-time biomedical signal processing on reconfigurable hardware, offering a practical and energy-efficient alternative to conventional deep learning approaches.

The key contributions and achievements of this project are summarized as follows:

- A complete neuromorphic ECG processing architecture was designed using the LIF neuronmodel, implemented in Verilog RTL, and deployed on the Zynq-7020 FPGA platform. The architecture consists of 10 LIF neurons with predefined synaptic weights, spike counting logic, and threshold-based abnormality detection.
- A hardware-software co-design approach was successfully implemented, integrating the ARM

  Cortex-A9 Processing System with the FPGA-based neuromorphic hardware accelerator through AXI DMA communication. This PS-PL integration enables efficient data transfer and seamless interaction between software control and hardware processing.
- The system was validated using real patient ECG signals from the MIT-BIH ArrhythmiaDatabase Record 100, demonstrating correct identification of abnormal ECG regions based on spike count analysis. The results show a strong correlation between spike activity patterns and ECG signal characteristics.
- The FPGA implementation achieves single-clock-cycle processing latency per sample throughparallel computation of all 10 neurons, demonstrating the realtime processing capability of the neuromorphic approach.
- The lightweight architecture utilizes minimal FPGA resources, leaving substantial capacity for

  future extensions and demonstrating the scalability of the proposed design.

The experimental results validate that the proposed FPGA-based neuromorphic system can effectively detect ECG abnormalities using spike-based computation. The system processes ECG signals in real time with low latency and minimal power consumption, making it suitable for edge computing and embedded healthcare applications. The successful demonstration of the end-to-end pipeline from dataset loading to hardware processing and result visualization confirms the practical applicability of the proposed approach.

In conclusion, this project bridges the gap between neuromorphic computing research and practical biomedical applications by demonstrating a fully functional SNN-based ECG abnormality detection system on commercially available FPGA hardware. The work establishes a strong foundation for future advancements in neuromorphic healthcare systems and contributes to the growing body of knowledge in hardware-accelerated biomedical signal processing.

6.2 FUTURE WORK

While the proposed system demonstrates the feasibility of neuromorphic ECG processing on FPGA,

several avenues for improvement and extension have been identified for future work:

Weight Optimization and Training: The current implementation uses predefined synaptic weights for hardware verification purposes. Future work will focus on implementing weight training mechanisms using neuromorphic learning rules such as Spike-Timing-Dependent Plasticity (STDP) or surrogate gradient methods. Training the weights on the MIT-BIH dataset will significantly improve the classification accuracy and enable the system to learn optimal feature representations from the ECG signals. The trained weights can be loaded into the weight ROM module for deployment.

Multi-Class Arrhythmia Detection: The current system performs binary classification (normal vs. abnormal). Future extensions will expand the classification capability to support multi-class arrhythmia detection, enabling the system to distinguish between specific types of cardiac abnormalities such as premature ventricular contractions (PVC), atrial fibrillation (AF), bundle branch blocks, and other arrhythmia types. This will require increasing the number of neurons and implementing more sophisticated classification logic.

Adaptive Threshold Optimization: The abnormality detection threshold is currently set as a fixed baseline value. Future work will investigate adaptive threshold mechanisms that can automatically adjust based on the patient-specific ECG characteristics and the trained weight configuration. Machine
Learning-based threshold optimization can further improve the detection sensitivity and specificity.

On-Chip Learning: Implementing on-chip learning capabilities will enable the system to adapt to individual patient ECG patterns over time, improving personalized cardiac monitoring. STDP-based learning rules can be implemented directly in the FPGA hardware, allowing the synaptic weights to be updated in real time based on the incoming ECG data.

Multi-Lead ECG Processing: The current system processes single-lead ECG signals. Future work will extend the architecture to support multi-lead ECG processing (e.g., 12lead

ECG), which provides more comprehensive cardiac information and enables more accurate diagnosis of complex cardiac conditions.

Physical ECG Sensor Integration: Integrating a physical ECG sensor with the PYNQ-Z2 board will enable live signal acquisition and real-time processing, moving the system from a validation prototype to a practical cardiac monitoring device. The Pmod connectors on the board can be used to interface with analog ECG front-end circuits.

Wearable Device Deployment: The ultimate goal is to deploy the neuromorphic ECG processing system on a miniaturized wearable device for continuous cardiac monitoring. This will require further optimization of the hardware design for power consumption, form factor reduction, and integration with wireless communication modules for remote monitoring.

Advanced Neuron Models: Exploring more complex neuron models such as the Adaptive Exponential Integrate-and-Fire (AdEx) model or the Izhikevich model may improve the classification performance by capturing more complex temporal dynamics in the ECG signals. These models can be implemented in Verilog RTL with moderate additional hardware resources.

Larger SNN Architectures: Scaling the SNN to include multiple layers and a larger number of neurons will enable the system to learn more complex feature hierarchies and improve classification accuracy. The available FPGA resources on the Zynq-7020 device can support significantly larger networks than the current 10-neuron implementation.

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APPENDIX

top_neuromorphic_ecg.v

```
`timescale 1ns / 1ps
  module top_neuromorphic_ecg(
    input  wire clk,
    input  wire signed [11:0] ecg_sample,
    output wire abnormal
);
 wire spike;
    wire fire;
   // Spike Encoder
    spike_encoder u_spike (
        .ecg_sample(ecg_sample),
        .spike(spike)
    );

    // LIF Neuron
    lif_neuron u_lif (
        .clk(clk),
        .spike_in(spike),
        .fire(fire)
    );
 // Abnormal Detector
    abnormal_detector u_abn (
        .clk(clk),
        .fire_in(fire),
        .abnormal(abnormal)
    );

endmodule.
```

spike_encoder.v

```
`timescale 1ns / 1ps
module spike_encoder #(
    parameter THRESHOLD = 200
)(
    input  wire signed [11:0] ecg_sample,
    output reg spike
);
 always @(*) begin
        if (ecg_sample > THRESHOLD)
            spike = 1'b1;
        else
            spike = 1'b0;
    end
endmodule
```

lif_neuron.v

```
`timescale 1ns / 1ps
module lif_neuron #(
    parameter THRESHOLD = 5,
    parameter LEAK      = 1
)(
    input  wire clk,
    input  wire spike_in,
    output reg  fire
```

```
);
  integer membrane = 0;
always @(posedge clk) begin
        // leak
        if (membrane > 0)
            membrane <= membrane - LEAK;
  // integrate
        if (spike_in)
            membrane <= membrane + 1;
    // fire condition
        if (membrane >= THRESHOLD) begin
            fire <= 1'b1;
            membrane <= 0;  // reset after fire
        end else begin
            fire <= 1'b0;
        end
    end
endmodule
```

abnormal_detector.v

```
`timescale 1ns / 1ps
module abnormal_detector #(
    parameter integer WINDOW_SIZE = 50,
    parameter integer FIRE_TH     = 2
)(
    input  wire clk,
```

```
    input  wire fire_in,
    output reg  abnormal
);
  integer fire_count  = 0;
    integer sample_count = 0;
 always @(posedge clk) begin
        // increment counters
        sample_count = sample_count + 1;
    if (fire_in)
            fire_count = fire_count + 1;
 // default output
        abnormal <= 1'b0;
  // window check
        if (sample_count >= WINDOW_SIZE) begin
            if (fire_count >= FIRE_TH)
                abnormal <= 1'b1;
    // reset window
            sample_count = 0;
            fire_count   = 0;
        end
    enD
endmodule
```

tb_ecg_reader.v

```
`timescale 1ns / 1ps
module tb_ecg_reader;
```

```
  reg clk;
  reg signed [11:0] ecg_sample;
  wire abnormal;
  integer i;
 // DUT
  top_neuromorphic_ecg dut (
      .clk(clk),
      .ecg_sample(ecg_sample),
      .abnormal(abnormal)
  );
 // clock
  always #5 clk = ~clk;
initial begin
      clk = 0;
 // simulate ECG waveform
      for (i = 0; i < 100; i = i + 1) begin
          ecg_sample = $random % 2000 - 1000;
#10;
$display("Sample %0d = %0d | Abnormal=%b",
                 i, ecg_sample, abnormal);
      end
$finish;
  end

endmodule
```